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Portfolio

BRUCE DAVIS

AI ENGINEER · UX / DESIGN ENGINEER

> Designer who codes and engineer who designs. Fifteen years shipping systems in regulated environments — identity platforms for 16,000+ users in HIPAA-compliant healthcare, payment infrastructure for SWIFT/CHIPS/Fedwire, and Python automation for **Non-Human Identity (NHI) governance at Microsoft** — the same identity problem that now defines enterprise AI agent deployment. Now applying that instinct to AI products: Claude-powered workflows, autonomous agents, MCP servers, document pipelines, and AI-native interfaces in React and TypeScript. Security, identity, and compliance as a feature, not a footnote.

RECENT AI WORK

SENTINEL TRIAGE

Autonomous SOC triage agent with a human-in-the-loop gate

A ReAct agent on Gemini 2.5 Flash that triages network security incidents — scans logs, enriches threat intelligence, proposes containment — through function-calling tools. Destructive actions pause for explicit human approval; an autonomy dial scopes how far the agent can act. Python · Streamlit · Gemini 2.5 Flash · google-genai SDK · function calling.

VISIONARY PRODUCER

Film pre-production orchestrator — one agent architecture, a new domain

A ReAct agent that turns raw script pages into a structured pre-production package — scene breakdown, VFX complexity scoring, casting archetypes — gated by producer approval. Built on the identical agent loop as Sentinel Triage: proof the architecture travels across domains. Python · Streamlit · Gemini 2.5 Flash · google-genai SDK.

LINEAR MCP

MCP server for Linear, with a full evaluation harness

Six opinionated Claude tools for running an issue tracker — notification triage, issue scoping, velocity diagnosis, status writeups. Includes a 30-case evaluation harness, a documented threat model, and per-tool cost analysis. TypeScript · @modelcontextprotocol/sdk · Anthropic SDK · Linear GraphQL API · Zod · Vitest.

DOCUMENT PIPELINE

Claude tools, end-to-end document processing

MCP-based set of Claude tools that handle a document end to end — extraction, PII masking / deidentification, structured JSON outputs, validation, human-in-the-loop review, and token-cost tracking. The Claude Engineer JD, end to end. MCP SDK · Node · ajv · JSON Schema · Python.

MCP SERVER

Custom MCP server, five Claude tools

Custom MCP server with five Claude tools for content workflow — Notion-backed idea retrieval, palette suggestion, voice-checking, page scaffolding, vault capture. Hands-on MCP server engineering and Claude API integration. @modelcontextprotocol/sdk · Notion API · Zod · Node.

AI INTERFACE PATTERNS

Loading, empty, and error states for AI surfaces

Five working React/TypeScript demos for AI products — streaming output, model state, low-context warnings, confidence flags, error recovery. Loading, empty, and error states built in code — not Figma mockups.

WORK

Business Execution Consultant — Infrastructure Design

Nov 2024–Apr 2025

Wells Fargo · Contract

- Mapped network dependencies for SWIFT, CHIPS, and Fedwire payment infrastructure — documented system flows that hadn't existed on paper, accelerating regulatory approval and reducing design rework by 25%.
- Designed NIST-compliant documentation architecture for network security controls; became the cross-team reference layer for incident response.
- Translated executive payment-rail objectives into detailed execution plans for ISO 20022 interoperability and Federal Reserve compliance.

Cloud Solutions Architect

May–Nov 2024

Microsoft · Contract

- Designed and built Python automation on the Microsoft Graph API to extract and normalize identity data — transformed raw Entra ID Governance output into executive-ready dashboards and documentation.

- Architected an AI-assisted workflow MVP using Microsoft Copilot — designed, built, and presented to global delivery stakeholders to accelerate knowledge transfer and standardize requirements gathering.
- Translated Non-Human Identity (NHI) governance capabilities into customer-facing solution designs — the same identity governance problem that now defines secure enterprise AI agent deployment.

Senior Security Engineer
Apr–Sept 2023

Starbucks · Contract

- Engineered PowerShell automation that replaced manual remediation work across 60+ domains — shipped scripts and runbooks at the engineering layer.
- Closed 50+ vulnerabilities per sprint through automation and scripted remediation rather than manual triage.
- Hardened identity and access controls across distributed teams; owned security testing and defect remediation end to end.

Project Manager — Infrastructure & IAM Delivery
Oct 2021–Nov 2022

Bank of America · Contract

- Led zero-downtime infrastructure cutover for 45+ engineering teams across Oracle, Azure, and Middleware platforms — coordinated dependency mapping, change validation, and stakeholder alignment.
- Owned ServiceNow dashboards as operational source of truth for change approvals, release readiness, and migration tracking; ensured all initiatives met stringent audit requirements.
- Authored runbooks and governance documentation for senior leadership; identified delivery risks through deep-dive requirements workshops.

Program Lead — IAM & Compliance
Jan 2019–Jul 2021

Veradigm · Healthcare

- Architected and shipped an IAM platform for 16,000+ users in a HIPAA-regulated healthcare environment — Entra ID, MIM, and CyberArk integrated end to end.
- Built Power Platform dashboards that surfaced compliance state across the organization; moved audit compliance from 65% to 95% over the engagement.
- Designed and executed a multi-year IAM strategy in a regulated healthcare environment, overseeing technology selection and solution architecture.

Earlier: Business Systems Consultant · Wells Fargo (2017–2018) · IAM Project Consultant · Cognizant (2016–2017) · Support Escalation Engineer · Microsoft (2012–2015)

STACK

DESIGN & FRONTEND: Figma · React · TypeScript · Next.js · Tailwind · Radix UI · Storybook · HTML/CSS

AI & AUTOMATION: Claude · Claude Code · Anthropic SDK · Gemini · google-genai SDK · MCP SDK · OpenAI SDK · Vercel AI SDK · Microsoft Copilot · Cursor · v0 · Figma Make

MOTION & INTERACTION: GSAP · Framer Motion · Three.js · Canvas API · D3.js · Matter.js · Web Audio API · WebGL

LANGUAGES & TOOLS: JavaScript · TypeScript · Node.js · Python · Streamlit · PowerShell · SQL · GraphQL · Zod · Vitest · Git · Vercel

ENTERPRISE & SECURITY: SWIFT · CHIPS · Fedwire · ISO 20022 · Entra ID Governance · Non-Human Identity · HIPAA · NIST · Threat Modeling · CyberArk · MIM · ServiceNow · JIRA

CERTIFICATIONS & EDUCATION

Certified in Cybersecurity (CC)
ISC2

Project Management Professional (PMP)
Project Management Institute

Professional Scrum Master I (PSM I)
Scrum.org

M.S., Computer Graphics Technology
Purdue University

B.S., Computer Science
Lincoln University